William C. Dawn

Curriculum Vitae

Education

May 2022 **Doctor of Philosophy, Nuclear Engineering**, *North Carolina State University*, Raleigh, NC
Research on unstructured mesh neutron transport methods and multiphysics modeling using GPUs.
Developed MEZCAL computer program using MFEM framework and exascale computing.
Dissertation: *Multiphysics Modeling of Microreactors with Unstructured Mesh Neutron Transport and Exascale Computing Architectures*.
Advisor: Scott Palmtag.

May 2019 **Master of Science, Nuclear Engineering**, *North Carolina State University*, Raleigh, NC
Nuclear Engineering University Program (NEUP) Fellow.
Developed LUPINE finite element simulation suite to model Sodium-cooled Fast Reactors (SFRs).
Thesis: *Simulation of Fast Reactors with the Finite Element Method and Multiphysics Models*.
Advisor: Scott Palmtag.

May 2017 **Bachelor of Science, Nuclear Engineering**, *North Carolina State University*, Raleigh, NC
Graduated as Valedictorian and Summa Cum Laude.
Led senior design team. Responsible for project management and interfacing with corporate sponsor.
Senior design project: "A Thermal Test Location in a Sodium Cooled Fast Reactor"

Professional Experience

June 2026 - Present **Computational Scientist**, *Idaho National Laboratory*, Idaho Falls, ID
- Developer in Computational Frameworks group.
- Finite Element Methods (FEMs) and High Performance Computing (HPC).

June 2022 - June 2026 **Senior Nuclear Engineer**, *Studsvik Scandpower Inc.*, Idaho Falls, ID
- Peacock lead developer.
  - New, continuous-energy Monte Carlo code developed from scratch within NQA-1 program.
  - Developed in C++23 with modern software best-practices (e.g., GitLab version control, unit tests, continuous integration, etc.).
  - Implemented relevant neutron physics including thermal scattering, unresolved resonances, and depletion.
  - Leading a team of five developers for nuclear data processing and input/output processing.
  - Re-established internship program and mentored interns.
- SIMULATE5/SIMULATE5-K developer.
  - Added capabilities for modeling Pellet-Cladding Mechanical Interaction (PCMI) during transients to support NRC Regulatory Guide 1.236.
  - Modeling of SPERT-III E-Core experiments and Ringhals BWR stability benchmark.
  - Implemented and benchmarked SIMULATE5-K-VVER for modeling transients in VVER-1000s and VVER-440s.

May 2019 - August 2019 **NEUP Intern**, *Idaho National Laboratory*, Idaho Falls, ID
- Added fast neutron cross section library capabilities to Griffin (Rattlesnake) neutron transport code based on MC^2-3 and ISOTXS file format.
- Used Griffin (Rattlesnake) to investigate effects of anisotropic neutron scattering in fast reactors.
- Implemented isotopic fission spectrum mixture into Griffin (Rattlesnake).
- Authored two technical reports related to work:
  - "Comparison of Higher-Order Neutron Scattering Cross Sections" (INL/EXT-19-54899)
  - "An Analytic Benchmark for the Solution to the Isotopic Fission Spectrum Mixture Problem" (INL/EXT-19-54998).

May 2018 - August 2019 **CASL Graduate Assistant**, *Consortium for Advanced Simulation of LWRs* (*CASL*), Raleigh, NC
- Developed lesson and lectured on CTF–MPACT coupling and use for practical reactor designs.
- Provided technical experience and IT support for student reactor design simulations.
- Contributed to logistical planning for CASL Institute.

May 2017 - August 2017 **NESLS Engineering Intern**, *Oak Ridge National Laboratory*, Oak Ridge, TN
- Added fast neutron cross section library capabilities to MPACT neutron transport code via ISOTXS file reader.
- Simulated fast neutron chloride molten salt reactor in steady-state and depletion simulations.
- Developed molten salt reactor models in MPACT, MCNP, and Serpent.

August 2015 - May 2017 **CASL Undergraduate Research Scholar**, *Consortium for Advanced Simulation of LWRs* (*CASL*), Raleigh, NC
- Reduced computing time by 30 % by improving steam tables in CTF.
- Performed code comparisons with MCNP to verify energy group structure in MPACT.

May 2016 - August 2016 **Edison Engineering Intern**, *GE Hitachi Nuclear Energy LLC*, Wilmington, NC
- PRISM
  - Drafted and submitted journal article "PRISM Reference Fuel Design."
  - Awarded two patents relating to ESBWR and one patent related to PRISM.
  - Developed PRISM General Description Book and prepared public-facing documents describing PRISM for a general audience.

May 2015 - August 2015
- LOCA & Containment
  - Analyzed reactor transients using TRACG to support 10 % power uprate.
  - Created automated data visualization and animation packages using MATLAB.

August 2014 - May 2017 **Licensed Reactor Operator**, *NCSU PULSTAR Research Nuclear Reactor*, Raleigh, NC
- Licensed by NRC to operate all controls at NCSU reactor facility.
- Experienced in startup, operation, and troubleshooting on 1 MW research reactor.

Publications

Dawn, William C., Gerardo Grandi, and Tamer Bahadir. "Validation of SIMULATE5-K and CASMO5 with the SPERT-III E-Core." In: *Frontiers in Nuclear Engineering* (2026).

Dawn, William C. "Method of Manufactured Solutions for the Neutron Kinetics Equations." In: *Nuclear Science and Engineering* (2025).

Dawn, William C. "A Benchmark for Neutron Kinetics Methods in Hexagonal Geometry." In: *Nuclear Science and Engineering* 199 (5 Sept. 2024).

Dawn, William C., Gerardo Grandi, and Tamer Bahadir. "Development and Benchmarking of SIMULATE5-K-VVER." In: *Annals of Nuclear Energy* 195 (Jan. 2024).

Dawn, William C. and Scott Palmtag. "Solving the Neutron Transport Equation for Microreactor Modeling Using Unstructured Meshes and Exascale Computing Architectures." In: *Nuclear Science and Engineering* 197.12 (Feb. 2023).

Dawn, William C. and Scott Palmtag. "A Multiphysics Simulation Suite for Liquid Metal-Cooled Fast Reactors." In: *Annals of Nuclear Energy* 159 (Sept. 2021).

Conference Papers

Wemple, Charles A., William Madsen, and **William C. Dawn**. "Benchmark Testing of the Peacock Monte Carlo Code System." In: *Proceedings of PHYSOR 2026*. Accepted. Torino, Italy, Apr. 2026.

Madsen, William and **William C. Dawn**. "Analysis of the Neutronic Effects of the Axial Twist of Helical Cruciform Fuel Rods Using Peacock." In: *Proceedings of PHYSOR 2026*. Accepted. Torino, Italy, Apr. 2026.

Dawn, William C., William Madsen, and Charles Wemple. "Investigation of Chlorine Cross Sections in Molten Chloride Fast Reactors using Peacock." In: *Proceedings of PHYSOR 2026*. Accepted. Torino, Italy, Apr. 2026.

Wemple, Charles A., William Madsen, and **William C. Dawn**. "Benchmark Testing of the Peacock Monte Carlo Code System." In: *Proceedings of PHYSOR 2026*. Accepted. Torino, Italy, Apr. 2026.

Nguyen, Tung D. C., **William C. Dawn**, Charles A. Wemple, Tamer Bahadir, Rodolfo M. Ferrer, Joshua Hykes, and Joel Rhodes III. "Multi-Group Cross Section Generation with Peacock for Fast Reactor Applications." In: *Proceedings of PHYSOR 2026*. Accepted. Torino, Italy, Apr. 2026.

Dawn, William C. and Charles Wemple. "Peacock: A Monte Carlo Transport Code with Depletion for Reactor Modeling." In: *Proceedings of Advances in Nuclear Fuel Management (ANFM) 2025*. Clearwater Beach, FL, July 2025.

Wemple, C. A., R. M Ferrer, T. T. Simeonov, J. M. Hykes, **W. C. Dawn**, and J. D. Rhodes. "Nuclear Data Usage at Studsvik Scandpower and Future Needs from ENDF/B-IX." In: *Proceedings of Advances in Nuclear Fuel Management (ANFM) 2025*. Clearwater Beach, FL, July 2025.

Ferrer, Rodolfo, **William C. Dawn**, and Tamer Bahadir. "Verification of CMS5 for Higher Enrichment and High Burnup PWR Core Design." In: *Proceedings of Advances in Nuclear Fuel Management (ANFM) 2025*. Clearwater Beach, FL, July 2025.

Dawn, William C. "Peacock: A Monte Carlo Code with Shared Memory Parallelism for Consumer-Grade Computing Architectures." In: *Proceedings of M&C 2025*. Denver, CO, Apr. 2025.

Dawn, William C., Charles Wemple, Joshua Hykes, Rodolfo M. Ferrer, and Joel Rhodes III. "Peacock: Development of a New Continuous Energy Monte Carlo Transport Code." In: *Transactions of the American Nuclear Society*. Vol. 131. Nov. 2024.

Dawn, William C., Gerardo Grandi, and Tamer Bahadir. "Initial Validation of SIMULATE5-K and the CMS5 Reactor Modeling Suite with the SPERT-III Experiments." In: *Proceedings of ANS PHYSOR 2024*. San Fransisco, CA, Apr. 2024.

Dawn, William C. and Tamer Bahadir. "Development and Benchmarking of Transient Nodal Code SIMULATE5-K Neutron Kinetics Solver for VVERs and Hexagonal Geometries." In: *Proceedings of ANS M&C 2023*. Niagara Falls, Ontario, Aug. 2023.

Kiefer, Timothy M., **William C. Dawn**, Khaldoon Al-Dawood, and Scott Palmtag. "Control Rod Modeling in Liquid Metal-Cooled Fast Reactors." In: *Proceedings of PHYSOR 2022*. Pittsburgh, PA, 2022.

Dawn, William C. and Scott Palmtag. "Simplified Thermal Expansion Modeling for Liquid Metal-Cooled Fast Reactors." In: *Proceedings of ANS M&C 2021*. Raleigh, NC, Oct. 2021.

Al-Dawood, Khaldoon A., **William C. Dawn**, and Scott Palmtag. "Multiphysics Simulation of Uranium-Nitride Fueled Lead-Cooled Fast Reactor." In: *Proceedings of ANS M&C 2021*. Raleigh, NC, 2021.

Palmtag, Scott, **William C. Dawn**, and Chase Lawing. "Fast Reactor Depletion Methods in LUPINE." In: *Proceedings of ANS M&C 2021*. Raleigh, NC, 2021.

Dawn, William C. and Scott Palmtag. "A Multiphysics Simulation Suite for Sodium Cooled Fast Reactors." In: *Proceedings of PHYSOR 2020* (Mar. 27–Apr. 6, 2020). Cambridge, UK, Apr. 2020.

Technical Reports

Dawn, William C., Javier Ortensi, Mark D. Dehart, and Scott P. Palmtag. *Comparison of Higher-Order Neutron Scattering Cross Sections*. Tech. rep. INL/EXT-19-54899. Idaho National Laboratory, 2019.

Dawn, William C. *An Analytic Benchmark for the Solution to the Isotopic Fission Spectrum Mixture Problem*. Tech. rep. INL/EXT-19-54998. Idaho National Laboratory, 2019.

Patents

Loewen, Eric P., James P. Sineath, Dean D. Molinaro, **William C. Dawn**, Robin D. Sprague, Theron D. Marshall, and Joel P. Melito. "Intermixing Feedwater Sparger Nozzles and Methods for Using the Same in Nuclear Reactors." 20180277265 (Wilmington, NC). Feb. 2020.

Loewen, Eric P., James P. Sineath, Dean D. Molinaro, **William C. Dawn**, William J. Garcia, Oscar L. Meek, and Patrick K. Day. "Acoustic Flowmeter and Methods of Using Same." 20180277267 (Wilmington, NC). May 2020.

Sineath, James P., Dean D. Molinaro, **William C. Dawn**, and Eric P. Loewen. "Systems and Methods for Airflow Control in Reactor Passive Decay Heat Removal." US10937557B2 (Wilmington, NC). Mar. 2021.

Technical Skills

Programming Languages	C++, Fortran, C, Python, MATLAB, LaTeX.	Simulation Packages:	MFEM, CASMO, SIMULATE, DIF3D, MC^2-3, REBUS, MCNP, Serpent, VERA, MPACT, CTF.
General Proficiencies	Bash scripting, git, GitHub & GitLab Project Management.		

Professional Development and Achievements

2021	Alan F. Henry/Paul A. Greebler – American Nuclear Scholarship.
2021	American Nuclear Society Mathematics & Computation Conference – Student Program Committee Co-Chair.
2019	NCSU College of Engineering Master's Scholar of the Year.
2017-2020	Nuclear Engineering University Program (NEUP) Fellowship. ○ Full funding for three years of graduate school. ○ Nationally recognized Department of Energy (DOE) fellowship.
2014-2017	Nuclear Engineering University Program (NEUP) Undergraduate Scholarship.
2014-2018	American Nuclear Society Scholarship.
2012	Eagle Scout.